

Descriptive and Frequentist Statistics

Module 3: Descriptive Statistics and Statistical Estimation

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Descriptive Statistics

Exploring the data

- Histogram
- Sample mean and variance
- Order statistics

Measuring dependence

- Sample covariance
- Sample covariance matrix

Frequentist Statistics

Foundations

- Sampling
- Mean square error
- Consistency

Inference and estimation

- Confidence intervals
- Parametric and non-parametric model estimation

Descriptive and Frequentist Statistics

Descriptive Statistics

- Summarizes the data that have been observed.
- Uses graphs and numerical measures to describe patterns.
- Examples: histogram, mean, variance and covariance.

Frequentist Statistics

- Uses repeated sampling to study unknown population quantities.
- Treats the parameter as fixed but unknown.
- Uses sampling distributions and estimators for inference.

Observed Data $\longrightarrow$ Describe Infer $\longrightarrow$ Population / Parameter

Population, Sample, Parameter and Statistic

- **Population:** Complete set of units or outcomes of interest.
- **Sample:** Subset observed from the population.
- **Parameter:** Numerical characteristic of the population.
- **Statistic:** Numerical quantity calculated from the sample.

Population $\longrightarrow$ Sampling $\longrightarrow$ Sample $\longrightarrow$ Statistic $\longrightarrow$ Parameter

Example

μ = Population mean

$\bar{X}$ = Sample mean

Parameter versus Statistic

Parameter

- Describes the **population**.
- Usually **unknown**.
- Represents the **target quantity**.

$$\mu, \sigma^2, p$$

Statistic

- Computed from the **sample**.
- Usually **observable**.
- Can be used as an **estimator**.

$$\bar{X}, S^2, \hat{p}$$

Key Idea

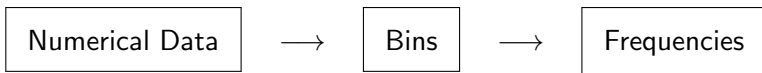
A statistic can be used to estimate a population parameter.

Histogram: Definition and Purpose

Definition

A **histogram** is a graphical representation of a quantitative variable. The observed values are divided into intervals called **bins**, and the height of each bar represents the frequency or relative frequency of observations.

- Displays the distribution of numerical data.
- Helps identify the **centre** and **spread** of the data.
- Reveals features such as **skewness** and **modality**.
- Can highlight unusual or extreme observations.



Construction of a Histogram

For observations $x_1, \dots, x_n$:

- 1 Find the minimum and maximum values.
- 2 Choose the number of bins or the bin width.
- 3 Define non-overlapping intervals.
- 4 Count the observations in each interval.
- 5 Plot the frequencies or densities against the intervals.

Histogram Density

If a bin has width h and contains f_j observations, its relative density can be written as

$$\hat{f}_j = \frac{f_j}{nh}$$

where n is the total number of observations.

- For equal-width bins, bar heights may represent frequencies.
- Using density makes the histogram area approximately sum to one.

Histogram: Frequency and Density Relation

- **Frequency:** Number of observations in a bin.
- **Relative frequency:** $\frac{f_j}{n}$
- **Density:** Relative frequency divided by the bin width.

For Equal-Width Bins

$$\text{Density} = \frac{\text{Frequency}}{n \times \text{Bin Width}}$$

- For unequal-width bins, the **bar area**, rather than height alone, represents probability.
- A density histogram has total area approximately equal to 1.

Histogram: Working Example

Consider the following observations representing test scores:

{42, 45, 47, 51, 53, 54, 56, 58, 61, 63, 65, 67, 68, 72, 75}

Step 1: Choose Bins

Using intervals of width 10:

Score Range	Frequency
40 – 49	3
50 – 59	5
60 – 69	5
70 – 79	2

Interpretation

Most observations lie between **50 and 69**, while relatively few observations occur at the lower and upper ends.

Histogram: Shape, Bin Width, Advantages and Limitations

Shape of a Histogram

- **Symmetric:** similar pattern on both sides.
- **Right-skewed:** longer tail to the right.
- **Left-skewed:** longer tail to the left.
- **Unimodal / Multimodal:** one or multiple peaks.

Advantages

- Simple visual summary of numerical data.
- Shows centre, spread and distributional shape.
- Helps identify skewness and unusual observations.
- Useful for exploratory data analysis.

Bin Width

- **Small bins:** reveal more detail but may appear irregular.
- **Large bins:** give a smoother view but may hide features.
- Bin width affects the level of detail visible in the histogram.

Limitations

- Appearance depends on bin width and bin boundaries.
- Different choices can give different visual impressions.
- Exact individual observations are not visible.

Sample Mean

For a sample of observations

$$X_1, X_2, \dots, X_n,$$

the **sample mean** is the arithmetic average of the observed values.

Formula

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- n denotes the sample size.
- $\bar{X}$ represents the sample mean.
- It measures the **central tendency** of the sample.
- The sample mean is commonly used to estimate the population mean μ .

Sample Mean as an Optimization Solution

The sample mean minimizes the sum of squared deviations from a constant value a .

$$\hat{a} = \arg \min_a \sum_{i=1}^n (X_i - a)^2$$

Differentiating with respect to a :

$$\frac{d}{da} \sum_{i=1}^n (X_i - a)^2 = -2 \sum_{i=1}^n (X_i - a)$$

Setting the derivative equal to zero:

$$\sum_{i=1}^n X_i - na = 0$$

Therefore,

$$a = \bar{X}$$

The sample mean is the **least-squares centre** of the observations.

Properties of the Sample Mean

For a random sample $X_1, X_2, \dots, X_n$ with population mean μ :

Key Properties

- The sample mean $\bar{X}$ is a statistic calculated from the observed sample.
- It is an **unbiased estimator** of the population mean:

$$E[\bar{X}] = \mu$$

- Its variability decreases as the sample size increases.
- For independent observations, the sample mean is based on information from all observations in the sample.

Variance of the Sample Mean

Let $X_1, X_2, \dots, X_n$ be independent observations from a population with variance σ^2 .

The variance of the sample mean is

Sampling Variance

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

- σ^2 is the population variance.
- n is the sample size.
- The variance of $\bar{X}$ decreases as n increases.
- Larger samples therefore produce sample means with less sampling variability.

Standard Error of the Sample Mean

The **standard error** measures the standard deviation of the sampling distribution of the sample mean.

Standard Error

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

- σ is the population standard deviation.
- n is the sample size.
- It measures the variability of $\bar{X}$ across repeated samples.
- A smaller standard error indicates a more precise estimate of μ .

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \implies SE(\bar{X}) = \sqrt{\text{Var}(\bar{X})}$$

Effect of Sample Size

The sample size n affects the variability and precision of the sample mean.

Key Relationship

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

- As n increases, the variance of $\bar{X}$ decreases.
- The standard error also decreases as n increases.
- Larger samples therefore provide more precise estimates of the population mean.

$$n \uparrow \implies SE(\bar{X}) \downarrow \implies \text{Greater Precision}$$

Sample Variance and Standard Deviation

For a sample $X_1, X_2, \dots, X_n$ with sample mean $\bar{X}$, the **sample variance** measures the spread of the observations around the sample mean.

Sample Variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- S^2 measures the variability within the sample.
- The denominator $n - 1$ provides an unbiased estimate of the population variance σ^2 .
- The **sample standard deviation** is

$$S = \sqrt{S^2}$$

- Standard deviation is expressed in the same units as the original observations.

Sample Variance: Worked Example

Consider the sample

$$X = \{2, 4, 6, 8, 10\}, \quad \bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6.$$

X_i	2	4	6	8	10	
$(X_i - \bar{X})^2$	16	4	0	4	16	$\sum (X_i - \bar{X})^2 = 40$

Variance Calculation

Divide by n

$$\frac{40}{5} = 8$$

Divide by $n - 1$

$$S^2 = \frac{40}{5 - 1} = 10$$

- The sample variance uses $n - 1$ to obtain an **unbiased estimate** of the population variance.

$$S^2 = 10$$

$\Rightarrow$

$$S = \sqrt{10} \approx 3.16$$

Order Statistics

Order statistics are the observations arranged in **ascending order**.

For a sample

$$X_1, X_2, \dots, X_n,$$

let

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

denote the ordered observations.

Key Order Statistics

- $X_{(1)}$ is the **minimum**.
 - $X_{(n)}$ is the **maximum**.
 - The **median** is the middle order statistic.
 - Quartiles are also obtained from the ordered observations.
-
- Order statistics are useful for describing the position and spread of observations.

Quantiles and Interquartile Range

Quantiles divide ordered data into parts containing approximately equal proportions of observations.

Important Quantiles

- Q_1 — first quartile (25th percentile)
- Q_2 — second quartile (50th percentile, median)
- Q_3 — third quartile (75th percentile)

The **interquartile range (IQR)** measures the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

- Quantiles describe the position of observations in an ordered sample.
- IQR is less affected by extreme observations than the range.

Outlier Detection using IQR

The IQR can be used to identify observations that are unusually far from the central part of the data.

IQR Rule

$$IQR = Q_3 - Q_1$$

$$\text{Lower Bound} = Q_1 - 1.5(IQR)$$

$$\text{Upper Bound} = Q_3 + 1.5(IQR)$$

- Observations below the lower bound are potential **lower outliers**.
- Observations above the upper bound are potential **upper outliers**.
- The rule is based on the middle 50% of the data.

Sample Covariance

Definition

Sample covariance measures the **joint variability** between two variables X and Y .

For paired observations $(X_1, Y_1), \dots, (X_n, Y_n)$,

Sample Covariance

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

- $S_{XY} > 0$: X and Y tend to increase together.
- $S_{XY} < 0$: one tends to increase when the other decreases.
- $S_{XY} \approx 0$: little linear co-variation is observed.

Interpretation

The sign indicates the direction of joint variation.

Covariance: Interpretation and Independence

Interpretation

- $S_{XY} > 0$: X and Y tend to increase together.
- $S_{XY} < 0$: one variable tends to increase as the other decreases.
- $S_{XY} \approx 0$: little linear co-variation is observed.

Covariance and Independence

If X and Y are independent, then

$$\text{Cov}(X, Y) = 0$$

provided the covariance exists. However,

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent}$$

- Zero covariance indicates absence of linear co-variation, but does not rule out other forms of dependence.

Sample Covariance: Numerical Example

Consider the paired observations $\{(X_i, Y_i)\} = \{(1, 2), (2, 3), (3, 5), (4, 6)\}$, where $\bar{X} = 2.5$ and $\bar{Y} = 4$.

Covariance Calculation

X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	2	-1.5	-2	3
2	3	-0.5	-1	0.5
3	5	0.5	1	0.5
4	6	1.5	2	3

$$\sum_{i=1}^4 (X_i - \bar{X})(Y_i - \bar{Y}) = 3 + 0.5 + 0.5 + 3 = 7$$

$$S_{XY} = \frac{7}{4 - 1} = \frac{7}{3} \approx 2.33$$

Interpretation

Since $S_{XY} > 0$, X and Y show **positive co-variation**.

Covariance and Correlation

Both covariance and correlation describe the relationship between two variables.

Correlation Coefficient

The sample correlation coefficient is the standardized form of sample covariance:

$$r_{XY} = \frac{S_{XY}}{S_X S_Y}$$

- **Covariance** depends on the units of X and Y .
- **Correlation** is unit-free and lies between -1 and 1 .
- $r_{XY} > 0$: positive linear association.
- $r_{XY} < 0$: negative linear association.
- $r_{XY} \approx 0$: weak or no linear association.

Important

Correlation measures **linear association**; it does not imply causation.

Sample Covariance Matrix and Its Properties

For p variables, the sample covariance matrix is

$$S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

For two variables X and Y ,

$$S = \begin{bmatrix} S_X^2 & S_{XY} \\ S_{XY} & S_Y^2 \end{bmatrix}$$

Properties

- **Symmetric:** $S = S^T$.
- Diagonal elements are the sample variances.
- Off-diagonal elements are the sample covariances.
- The covariance matrix is positive semidefinite.

Covariance Matrix: Numerical Example

Consider two variables with sample variances

$$S_X^2 = 4, \quad S_Y^2 = 9$$

and sample covariance

$$S_{XY} = 3.$$

The sample covariance matrix is

$$S = \begin{bmatrix} S_X^2 & S_{XY} \\ S_{XY} & S_Y^2 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix}$$

Interpretation

- The diagonal elements 4 and 9 are the variances of X and Y .
- The off-diagonal elements 3 represent the covariance between X and Y .
- Since $S_{XY} > 0$, the variables show positive co-variation.

Covariance Matrix: Interpretation

For variables $X_1, X_2, \dots, X_p$, the covariance matrix summarizes their variances and pairwise covariances.

$$S = \begin{bmatrix} S_{X_1}^2 & S_{X_1X_2} & \cdots & S_{X_1X_p} \\ S_{X_2X_1} & S_{X_2}^2 & \cdots & S_{X_2X_p} \\ \vdots & \vdots & \ddots & \vdots \\ S_{X_pX_1} & S_{X_pX_2} & \cdots & S_{X_p}^2 \end{bmatrix}$$

Interpretation

- **Diagonal elements:** variances of individual variables.
- **Off-diagonal elements:** covariances between pairs of variables.
- Positive covariance indicates variables tend to vary in the same direction.
- Negative covariance indicates variables tend to vary in opposite directions.

PCA and Explained Variance

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms correlated variables into a smaller set of uncorrelated components.

Key Idea

PCA uses the **covariance matrix** to find directions of maximum variance in the data.

Covariance Matrix $\longrightarrow$ Principal Components $\longrightarrow$ Explained Variance

If λ_i is the eigenvalue of the i -th component,

$$\text{EVR}_i = \frac{\lambda_i}{\sum_{j=1}^p \lambda_j}$$

- Components are ordered by the amount of variance they capture.
- Cumulative explained variance helps decide how many components to retain.

Multivariate Gaussian Model

A multivariate Gaussian distribution extends the Gaussian distribution to multiple variables.

$$\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Key Parameters

- $\boldsymbol{\mu}$: mean vector
 - $\boldsymbol{\Sigma}$: covariance matrix
 - $\boldsymbol{\Sigma}$ controls the spread and orientation of the distribution.
-
- The covariance matrix captures the relationships between the variables.
 - It provides a model for the joint distribution of multivariate data.

Frequentist Statistics and Sampling

Frequentist Statistics

Frequentist statistics studies population parameters through the behaviour of statistics obtained from repeated samples.

- Population parameters are treated as **fixed but unknown**.
- Samples are drawn from the population to obtain information about these parameters.
- Different samples can produce different values of a statistic.

Population $\longrightarrow$ Sampling $\longrightarrow$ Sample $\longrightarrow$ Inference

Sampling Methods and Bias

Probability Sampling

- **Simple random:** every unit has equal probability of selection.
- **Systematic:** select every k th unit after a random start.
- **Stratified:** divide the population into strata and sample from each.
- **Cluster:** select naturally occurring groups or clusters.

Sampling Bias

Sampling bias occurs when some parts of the population are systematically over- or under-represented.

- Coverage bias
- Non-response bias
- Convenience or voluntary-response bias

Important

Increasing sample size does not automatically remove systematic bias.

Sampling Distribution of the Sample Mean

For independent observations with population mean μ and variance σ^2 ,

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

If the population is normally distributed,

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Central Limit Theorem

Even when the population is not normal, the sampling distribution of $\bar{X}$ becomes approximately normal for sufficiently large n .

- Larger samples produce a narrower sampling distribution.
- The sampling distribution forms the basis for statistical inference.

Mean Square Error (MSE)

The **Mean Square Error (MSE)** measures the average squared difference between an estimator and the true parameter.

For an estimator $\hat{\theta}$ of θ ,

$$MSE(\hat{\theta}) = E \left[(\hat{\theta} - \theta)^2 \right]$$

Bias-Variance Decomposition

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

where

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- Lower MSE indicates an estimator with smaller overall error.

Bias, Variance and Unbiasedness

For an estimator $\hat{\theta}$ of a parameter θ :

Bias

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- **Unbiased estimator:**

$$E[\hat{\theta}] = \theta$$

- **Variance:** measures the variability of the estimator across repeated samples.
- Lower variance means more stable estimates.

$$MSE = \text{Variance} + \text{Bias}^2$$

MSE: Machine Learning Example

In regression, MSE measures the average squared difference between the **actual** and **predicted** values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Consider

Actual (y_i)	3	5	7
Predicted ($\hat{y}_i$)	2	6	8

$$MSE = \frac{(3-2)^2 + (5-6)^2 + (7-8)^2}{3} = \frac{3}{3} = \boxed{1}$$

- Lower MSE indicates smaller prediction error.

Consistency of an Estimator

An estimator $\hat{\theta}_n$ is **consistent** for a parameter θ if it approaches the true parameter as the sample size increases.

Definition

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

This means that for any $\varepsilon > 0$,

$$P\left(|\hat{\theta}_n - \theta| > \varepsilon\right) \longrightarrow 0$$

- Larger samples make the estimator increasingly close to θ .
- Consistency is a desirable property of an estimator.

Consistency of the Sample Mean

For independent observations with finite population variance,

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

As the sample size increases,

$$\frac{\sigma^2}{n} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty$$

Therefore,

$$\boxed{\bar{X} \xrightarrow{P} \mu}$$

Conclusion

The sample mean is a **consistent estimator** of the population mean.

Confidence Intervals

A **confidence interval (CI)** gives a range of plausible values for an unknown population parameter based on sample data.

General Form

$$\text{Estimate} \pm \text{Margin of Error}$$

For a parameter θ ,

$$\hat{\theta} \pm (\text{Critical Value})(\text{Standard Error})$$

- The confidence level describes the long-run coverage of the interval procedure.
- Common confidence levels are 90%, 95%, and 99%.
- A larger sample generally gives a narrower interval.

Confidence Interval for the Population Mean

Known Population Standard Deviation

If σ is known,

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Unknown Population Standard Deviation

If σ is unknown, use the sample standard deviation S :

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

- $z_{\alpha/2}$ is used when σ is known.
- $t_{\alpha/2, n-1}$ is used when σ is unknown.

Confidence Interval: Worked Example

Suppose a sample of $n = 25$ observations has

$$\bar{X} = 50, \quad S = 5$$

Find the 95% confidence interval for the population mean μ . Since σ is unknown, use the t -distribution:

$$t_{0.025,24} \approx 2.064$$

Calculation

$$CI = \bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

$$= 50 \pm 2.064 \frac{5}{\sqrt{25}}$$

$$= 50 \pm 2.064$$

$$CI = (47.936, 52.064)$$

Confidence Interval Width and Population Proportion

Confidence Interval Width

For a population mean, interval width depends on variability, confidence level, and sample size.

$$\text{Width} \propto \frac{\text{Critical Value} \times \text{Standard Deviation}}{\sqrt{n}}$$

Greater variability / confidence $\Rightarrow$ wider interval; larger $n \Rightarrow$ narrower interval.

Population Proportion

For a sample proportion, $\hat{p} = x/n$, with approximate standard error

$$SE(\hat{p}) \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

For a large sample, the approximate confidence interval is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Parametric Model Estimation

Parametric estimation assumes that the population distribution belongs to a family described by a finite number of parameters.

Basic Idea

Sample Data $\longrightarrow$ Estimate Parameters $\longrightarrow$ Fitted Model

For example, a Gaussian model is described by

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

where μ and σ^2 are the unknown parameters.

- Parameters can be estimated using methods such as **Method of Moments** and **Maximum Likelihood Estimation**.

Method of Moments

The **method of moments** estimates unknown parameters by equating sample moments with the corresponding theoretical moments.

First Moment

For a population with mean μ , $E[X] = \mu$. The sample first moment is

$$\frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

For a model with parameters $\theta_1, \dots, \theta_k$, sample moments are matched with the corresponding theoretical moments.

$$\text{Sample Moments} = \text{Theoretical Moments}$$

- Solve the resulting equations to obtain parameter estimates.
- It provides a simple way to estimate parameters from sample data.

Maximum Likelihood Estimation

Maximum Likelihood Estimation (MLE) estimates the unknown parameter by choosing the value that makes the observed sample most likely. For observations $x_1, x_2, \dots, x_n$ with density or PMF $f(x; \theta)$, the likelihood function is

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

The maximum likelihood estimator is

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} L(\theta)$$

Log-Likelihood

Since products can be inconvenient, the log-likelihood is used:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$$

MLE of Normal Mean and Variance

Suppose the observations follow a Normal distribution:

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

Maximizing the likelihood gives

$$\hat{\mu}_{MLE} = \bar{X}$$

and

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

- $\hat{\mu}_{MLE}$ equals the sample mean.
- The variance MLE differs from the unbiased sample variance.

Non-Parametric Model Estimation

Non-parametric methods impose fewer assumptions about the functional form of the population distribution.

Key Idea

Instead of assuming a specific distribution such as Normal or Poisson, the distribution is estimated more flexibly from the data.

- **Empirical Distribution Function (ECDF)**
- **Kernel Density Estimation (KDE)**
- Rank-based methods

Advantages

- Flexible when distributional assumptions are uncertain.
- Useful for exploring the underlying distribution of data.

Trade-off

Greater flexibility can require more data and careful tuning.

Empirical Distribution Function and KDE

Empirical Distribution Function (ECDF)

The ECDF estimates the cumulative distribution directly from the observed sample:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq x)$$

It gives the proportion of observations less than or equal to x .

Kernel Density Estimation (KDE)

KDE provides a smooth estimate of the probability density:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

where K is the kernel function and h is the bandwidth.

Parametric vs Non-Parametric Estimation

Parametric	Non-Parametric
Assumes a specific distributional form	Makes fewer distributional assumptions
Lower flexibility	Higher flexibility
Often requires fewer data	Often requires more data
Often highly interpretable	Interpretability depends on the method
Normal MLE, Poisson	ECDF, KDE

Key Difference

Parametric estimation relies on a specified model with a finite set of parameters, whereas non-parametric estimation allows the distribution to be estimated more flexibly from the data.

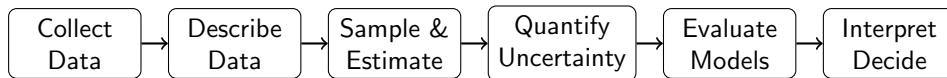
Statistics in Data Science and AI

- **Exploratory Data Analysis (EDA):** summarise data and identify patterns, spread and anomalies.
- **Feature Engineering:** identify redundant, variable and correlated features.
- **Dimensionality Reduction:** use covariance structure and PCA to retain important variation.
- **Machine Learning:** estimate model parameters and quantify prediction error.
- **Experimentation:** compare treatments, models or algorithms using statistical methods.

Role of Statistics

Statistics provides methods for understanding data, estimating unknown quantities, measuring uncertainty and supporting data-driven decisions.

Integrated Statistical Workflow



Overall Idea

Descriptive statistics summarize observed data, while frequentist methods use sampling and estimation to quantify uncertainty and draw conclusions about population parameters.

Key Takeaways

- **Descriptive statistics** summarise the main features of observed data using numerical and graphical methods.
- **Histograms, moments and order statistics** help describe the distribution, centre and spread of data.
- **Covariance and covariance matrices** describe relationships and joint variability among variables.
- **Frequentist statistics** uses repeated sampling to estimate unknown population parameters.
- **MSE, bias, variance and consistency** are important properties for evaluating estimators.
- **Confidence intervals** quantify uncertainty in parameter estimates.
- **Parametric and non-parametric estimation** provide different approaches depending on distributional assumptions.
- These statistical concepts form an important foundation for **Data Science and AI**.

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